Extending Our Page with Turbo Streams

Turbo Streams is the next step in using Turbo to make pages more dynamic without writing any custom JavaScript. I encourage you to think of this development process as progressive enhancement. We started with a working page that we thought needed to be more interactive. We added inline editing, we added more responsive updates to part of the page, and now we're going to add updating multiple parts of a page from a request.

Turbo Streams allows you to send an arbitrary amount of HTML content to the page and have that content replace or augment existing content in multiple arbitrary locations on the page. It's designed for exactly our situation, where we want multiple parts of the page to update from a single request. Turbo Streams is also designed to work really well with WebSockets via ActionCable, which I'll talk about later in Chapter 9, *Immediate Communication with ActionCable*.

A Turbo Stream is another custom HTML tag, `<turbo-stream>`, which has two attributes and a child tag of `template`. The attributes must include an `action` attribute and either a singular `target` or a plural `targets` attribute. Generically, a Turbo Stream element looks like this (an HTTP response can contain an arbitrary amount of Turbo Stream tags):

```
<turbo-stream action="<ACTION>" target="<TARGET>">
  <template>
    OUR HTML GOES HERE
  </template>
</turbo-stream>
```

The `target` or `targets` are the DOM ID of the element of the page being changed. The `action` specifies what to do with the incoming HTML. There are seven actions:

- `after`: The body of the template is added after the element with the target ID. The new code is outside the target ID element.
- `append`: The body of the template is added at the end of the `target`, and the existing contents are maintained.